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PI Name:	Dutta, Prabir Ph.D.		
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PI Email:	dutta.1@osu.edu	**Fax:**	FY 614-688-5402
PI Organization Type:	UNIVERSITY	**Phone:**	614-292-4532
Organization Name:	The Ohio State University		
PI Address 1:	Department of Chemistry		
PI Address 2:	100 West 18th Avenue		
PI Web Page:	http://www.chemistry.ohio-state.edu/		
City:	Columbus	**State:**	OH
Zip Code:	43210-1173	**Congressional District:**	15
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Contact Monitor:	Gilles, Donald	**Contact Phone:**	256 544-9302
Contact Email:	donald.C.Gillies@msfc.nasa.gov		
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Our goal is to assemble a supramolecular structure around a central zeolite membrane with the following features.
· Energy transfer from extrazeolitic molecules to sensitizer molecules on the surface of the zeolite membrane

· Charge (electron) transfer from the photoexcited sensitizer molecule to acceptor molecules within the zeolite membrane

· Charge transport through the membrane by electron hopping

· Catalytic oxidation and reduction of H2O to oxygen and hydrogen on two opposite sides of the membrane and regeneration of the donor and acceptor

· Zeolite pore engineering to ensure entrapment of photoactive molecules, but still allow for proton migration to

Task Description:	maintain charge balance
	
	The task is divided into four subcategories, dealing with zeolitic energy transfer, electron transfer processes on the surface of the zeolite, intrazeolitic charge propagation and zeolite-based nanostructured catalysts. Choice of molecules include polypyridyl complexes of Ru and Os as sensitizers and electron donors and bipyridinium-based systems as electron acceptors and metal oxides as catalysts. Photosynthesis, the conversion of sunlight to useful chemical energy is the basis for life. Important concepts that have emerged from studies on natural photosynthesis include the structure of the photoactive molecules, novel spatial arrangement, directed electron transfer and catalysis. Our objective is to design an artificial photosynthetic assembly with the goal of photolytic splitting of water into hydrogen and oxygen. We propose to use microporous crystalline, aluminosilicate zeolites as the host system. The challenge in design of complex, multicomponent functional assemblies from nanoscale building blocks is the optimization and assembly of the various components. Since the ordered array of channels and cages exist in nanocrystallites as well as in larger zeolite membranes, a convenient template for self assembly over large length scales becomes possible. Choice of molecules in this study include polypyridyl complexes of Ru and Os as sensitizers and electron donors and bipyridinium-based systems as electron acceptors. The nanometer size cages and channels of aluminosilicate zeolites will anchor the photochemical moieties as well as the catalysts necessary for reaction with charge separated redox species. The specific objectives of the research program includes:
	
	$ energy transfer within nanocrystalline (~20-100nm) zeolite crystallites, as well as energy transfer into and out of a zeolite
	
	$ electron transfer dynamics from surface-bound sensitizers into acceptor molecules in zeolites
	
	$ synthesis of zeolitic membranes from nanocrystalline zeolites and photochemical charge propagation within zeolite membranes
	
	$ use of nanometer sized zeolite cages for synthesis of metal oxide catalysts for multi-electron processes
	
	· assembly of the above components into a multifunctional unit capable of light-driven conversion of water to H2 and O2. Long term space exploration and habitation necessitate that systems be developed for generation of molecular building blocks that can be assembled to form complex macroscopic structures, such as fuel, polymers, drugs, etc. Ideally, the process should begin with the simplest of molecules that are readily available and use a plentiful energy source. The best example in nature of such a process is photosynthesis, which, in plants, involves the solar energy-driven catalytic reaction of making carbohydrates from water and carbon dioxide. This is the basis on which life is sustained on earth. The conversion of H2O and CO2 to O2 and carbohydrates is a complex redox process, involving a large array of molecules and a complex membrane-based architecture. In principle, many solar energy-based transformations are possible, in which simple molecules like H2O, CO2, N2 can be converted to molecular building blocks such as H2, CH4, NH3, which can form the basis for manufacture of important chemicals necessary for life.
	
	Solar radiation can drive these uphill reactions. The simplicity of these reactions masks the complexity of the architecture necessary to accomplish the goal. However, natural photosynthesis provides significant clues for assembly of "artificial" photosynthetic systems, and in this proposal, we propose a zeolite-based architecture, that at a primitive level mimics natural photosynthesis. For our model redox reactions, we have chosen the splitting of H2O into H2 and O2. The principles we develop in this program should be applicable to other photoredox systems, especially the architectural aspects related to assembly of multifunctioning, multicomponent units. The limitations of weight that can be carried on long term missions as well as space habitation require that materials taken from earth can serve several functions. Zeolites, because of their applications in adsorption, ion-exchange, separations and catalysis can play that role. In addition, photosensitized reactions for both environmental pollution control (air and water purification) as well as synthesis of fine chemicals have been reported with zeolites, and is currently satisfying aspects of “sustainable/green chemistry” on earth .

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Successful artificial photosynthesis systems that can manufacture fuels and chemicals using sunlight and simple chemicals such as water and carbon dioxide as feed will revolutionize our energy dependence. These technologies are examples of "disruptive technologies", since they set up a new paradigm of energy utilization.
	In the past several decades there has been considerable interest in efficiently harnessing solar energy to generate chemical energy. However, significant progress is still needed to reach this goal. Natural photosynthesis represents the conversion of solar energy to chemicals via a complex array of light-harvesting pigments and redox species arranged spatially across membranes. It is necessary to create long-lived accessible photogenerated charge-separated species for building up an efficient energy conversion system. The thermal back-electron-transfer reaction makes this difficult; e.g., with the photosensitizer tris(bipyridine)ruthenium-(II) (Ru(bpy)32+) and the electron acceptor methyl viologen (MV 2+), the forward- and back-electron-transfer rate constants are reported to be 5.6 x 10 9 and 2.4 x 10 9 M -1 s -1, respectively. Various heterogeneous supports have been examined for the Ru(bpy)3 2+ and viologen system such as micelles, vesicles, silica gel, clays, and zeolites. Zeolite-based systems have shown promise for generation of long-lived charge-separated species. Several groups have shown that the back electron transfer between zeolite-entrapped Ru(bpy)33+ and viologen radical can be slowed, and charge hopping between intrazeolitic viologen molecules can lead to permanent photoinduced charge separation. However, in these latter cases, Ru-(bpy)32+ was within the zeolite and not accessible for possible chemical utilization. Our focus has been on addressing the accessibility problem by examining a system in which a polypyridine-ruthenium moiety is covalently bonded to entrapped-bipyridinium acceptors on the surface of the zeolite. By adapting a zeolite membrane, which can keep apart two chemical systems that interfere with each other, yet still allow for transport of specific species through the membrane, a spatial separation of the photogenerated hole and electron can be achieved. In our study a prototype zeolite-membrane-based artificial photosynthetic assembly, which is a Ru polypyridyl complex decorated on the surface of the zeolite with intrazeolite bipyridinium ions and propylviologen sulfonate in solution, is proposed as an attractive feature in mimicking natural photosynthesis. An electron donor and acceptor dyad [Ru(bpy)2(L-4DQ)]4+-Y zeolite was chosen as a model system. [Ru(bpy)2(L-4DQ)]4+-Y ion-exchanged with MV2+ upon exposure to ambient light in an anaerobic environment exhibited new bands at 390 and 605 nm. Further, these spectral changes disappeared when the zeolite sample was subsequently exposed to O2. These observations are consistent with formation of the MV·- radical.

Task Progress:	Ru(bpy)2(L-4DQ)4+ -Y ion-exchanged with 3DQ2+ was dispersed in a PVS solution and irradiated with visible light. A growth in bands at 390 and 605 nm signifying the formation of PVS·- radical in solution was noted. Several experiments were done with the solution recovered from the photolysis sample and measured their excitation (measured at 530 nm) and emission spectrum (lmax: 530 nm). A new species was being created during photolysis with characteristic absorption and emission maximum at 390 and 530 nm indicated the presence of 3,4-dihydro-1,1'-dimethyl-3-oxo-4,4'-bipyridinium cation (3-one), which originates from be the oxidation of PVS in solution. The question that arises is what species oxidizes PVS to the pyridones? In aqueous solution at neutral pH, the half-life for Ru-(bpy)3 3+ is of the order of 200 s in the dark and is considerably faster in the presence of visible light. Illumination within the ligand to metal charge-transfer band of Ru(bpy)33+ increases the nucleophilicity of the bpy ligand and thereby promotes the attack by water. In earlier studies, we concluded that the covalent hydrates and hydroxylated species of the ruthenium complex formed by the attack of water on Ru(bpy)33+ oxidize methyl viologen to pyridones and regenerate Ru(bpy)3 2+. We propose that, in the [Ru(bpy)2(L-4DQ)]4+ -3DQ 2+ -zeolite sample, photoexcitation leads to electron transfer from the tethered Ru complex to intrazeolitic 3DQ2+. On the basis of data, we concluded that, in the [Ru(bpy)2(L-4DQ)]4+ -3DQ 2+ -zeolite sample, photoexcitation leads to electron transfer from the tethered Ru complex to intrazeolitic 3DQ 2+. The electron on the 3DQ 2+/+ can migrate through the zeolite and upon coming to the surface of the zeolite is vectorially transferred (because of the difference in reduction potentials) to PVS in solution forming PVS·- , a feature that has also been reported by other groups. Zeolite-based RuO2 catalytic systems that can result in O2 generation are currently being incorporated into the molecular assembly and will assist in regeneration of the Ru(II) complex. Here, it is suggested that using a zeolitic membrane to separate the redox species would alleviate this problem. A prototype zeolite-based artificial photosynthetic assembly is proposed for demonstrating efficient long-lived charge transfer and separation. The assembly will be configured to have Ru polypyridyl complex decorating the surface of the zeolite membrane with intrazeolitic bipyridinium ions (3-DQ2+). Its photochemical and photophysical properties will be examined in the same manner as with zeoltie particles and it is expected that a long-lived charge separation across a membrane will be possible. This successful demonstration should lead to interesting chemistry, including the possibility of splitting water into hydrogen and oxygen.
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